

KITABU QUEST

S6

CORE MATHEMATICS



Cover scene

students solving mathematics with graphs and models with a mountain gorilla mascot

● Clear lessons

● Worked examples

● Fun activities

● Real-life problems

GRADE

6

Title Page

KITABU QUEST S6 Core Mathematics

Rwanda REB-CBC learner book

Mascot: mountain gorilla

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Rwanda learners.

How To Use This Book

This book helps S6 learners study Core Mathematics one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Use models, diagrams, algebra, graphs, and worked examples before independent practice.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

1. Complex numbers
2. Logarithmic and exponential functions
3. Taylor and Maclaurin's expansions
4. Integration
5. Differential equations
6. Intersection and sum of subspaces
7. Transformation of matrices
8. Conics
9. Random variables

As you finish each topic, return to the success criteria and correct one answer before moving on.

Complex numbers: Lesson Opener

Learning focus: Complex numbers

Country link: a Rwanda school, market, farming, building, transport, or data problem for Complex numbers.

By the end of this topic, you should be able to:

- perform operations on complex numbers in different forms and use complex numbers to solve related problems in physics (voltage and current in alternating current), computer science (fractals), trigonometry (Euler's formula to transform trigonometric expression)
- identify the real part and the imaginary part of a complex number
- convert a complex number from one form to another
- represent a complex number on Argand diagram
- state De Moivre's formula and Euler's formula
- apply the properties of complex numbers to perform operations on complex numbers in algebraic form, in polar form or in exponential form

Inquiry questions:

- How can complex numbers help you solve a real problem in Core Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Complex numbers: Learn

Complex numbers teaches how two quantities change together at the same rate.

Core idea:

- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:

- Complex numbers
- Complex
- numbers
- method
- model
- unit
- answer
- check

Complex numbers: Worked Example

Worked example for Complex numbers: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: $900 / 3 = 300$ francs.
2. Multiply by 5 books: $300 \times 5 = 1500$ francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Complex numbers: Vocabulary And Study Skill

Use these words and study moves before you practise Complex numbers. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Complex numbers
- Complex
- numbers
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Complex numbers without a clear example, method, or evidence.

Complex numbers: Guided Activity

Guided activity for Complex numbers:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Complex numbers: Practice And Correction

Practice for Complex numbers:

Main outcome: perform operations on complex numbers in different forms and use complex numbers to solve related problems in physics (voltage and current in alternating current), computer science (fractals), trigonometry (Euler's formula to transform trigonometric expression).

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:

Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because $18 / 2 = 9$ km per hour.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to complex numbers and solve it.

Complex numbers: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Complex numbers.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Complex numbers: Outcome Check

Outcome check for Complex numbers: Perform operations on complex numbers in different forms and use complex numbers to solve related problems in physics (voltage and current in alternating current), computer science (fractals), trigonometry (Euler's formula to transform trigonometric expression).

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Complex numbers: Outcome Check

Outcome check for Complex numbers: Identify the real part and the imaginary part of a complex number.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Logarithmic and exponential functions: Lesson Opener

Learning focus: Logarithmic and exponential functions

Country link: a Rwanda school, market, farming, building, transport, or data problem for Logarithmic and exponential functions.

By the end of this topic, you should be able to:

- extend the concepts of functions to investigate fully logarithmic and exponential functions, finding the domain of definition, the limits, asymptotes, variations, graphs, and model problems about interest rates, population growth or decay, magnitude of earthquake
- state the restrictions on the base and the variable in a logarithmic function
- extend the concept of functions to investigate fully logarithmic and exponential functions
- perform operations on logarithmic
- transform a logarithm from a base to another
- find the domain and the range of a logarithmic or an exponential function

Inquiry questions:

- How can logarithmic and exponential functions help you solve a real problem in Core Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Logarithmic and exponential functions: Learn

Logarithmic and exponential functions teaches how two quantities change together at the same rate.

Core idea:

- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:

- Logarithmic and exponential functions
- Logarithmic
- exponential
- functions
- method
- model
- unit
- answer

Logarithmic and exponential functions: Worked Example

Worked example for Logarithmic and exponential functions: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: $900 / 3 = 300$ francs.
2. Multiply by 5 books: $300 \times 5 = 1500$ francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Logarithmic and exponential functions: Vocabulary And Study Skill

Use these words and study moves before you practise Logarithmic and exponential functions. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Logarithmic and exponential functions
- Logarithmic
- exponential
- functions
- method
- model
- unit

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Logarithmic and exponential functions without a clear example, method, or evidence.

Logarithmic and exponential functions: Guided Activity

Guided activity for Logarithmic and exponential functions:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Logarithmic and exponential functions: Practice And Correction

Practice for Logarithmic and exponential functions:

Main outcome: extend the concepts of functions to investigate fully logarithmic and exponential functions, finding the domain of definition, the limits, asymptotes, variations, graphs, and model problems about interest rates, population growth or decay, magnitude of earthquakes.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:

Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because $18 / 2 = 9$ km per hour.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to logarithmic and exponential functions and solve it.

Logarithmic and exponential functions: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Logarithmic and exponential functions.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Logarithmic and exponential functions: Outcome Check

Outcome check for Logarithmic and exponential functions: Extend the concepts of functions to investigate fully logarithmic and exponential functions, finding the domain of definition, the limits, asymptotes, variations, graphs, and model problems about interest rates, population growth or decay, magnitude of earthquakes.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Logarithmic and exponential functions: Outcome Check

Outcome check for Logarithmic and exponential functions: State the restrictions on the base and the variable in a logarithmic function.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Taylor and Maclaurin's expansions: Lesson Opener

Learning focus: Taylor and Maclaurin's expansions

Country link: a Rwanda school, market, farming, building, transport, or data problem for Taylor and Maclaurin's expansions.

By the end of this topic, you should be able to:

- use Taylor and Maclaurin's expansion to solve problems about approximations, limits and integration. Extend the Maclaurin's expansion to Taylor series
- state the Taylor's formula for a function and the conditions for the expansion to be valid
- state the Maclaurin's formula for a function and the conditions for the expansion to be valid
- list the first few terms of Maclaurin's expansion of $(1+x)^n$, $\cos x$, $\sin x$, e^x and $\ln(1+x)$, in each case, state the conditions for the expansion to be valid
- derive the Maclaurin's series for $(1+x)^n$, $\cos x$, $\sin x$, e^x and $\ln(1+x)$
- use Maclaurin's series to approximate an irrational number or the roots of a transcendental equation

Inquiry questions:

- How can Taylor and Maclaurin's expansions help you solve a real problem in Core Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Taylor and Maclaurin's expansions: Learn

Taylor and Maclaurin's expansions teaches how two quantities change together at the same rate.

Core idea:

- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:

- Taylor and Maclaurin's expansions
- Taylor
- Maclaurin's
- expansions
- method
- model
- unit
- answer

Taylor and Maclaurin's expansions: Worked Example

Worked example for Taylor and Maclaurin's expansions: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: $900 / 3 = 300$ francs.
2. Multiply by 5 books: $300 \times 5 = 1500$ francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Taylor and Maclaurin's expansions: Vocabulary And Study Skill

Use these words and study moves before you practise Taylor and Maclaurin's expansions. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Taylor and Maclaurin's expansions
- Taylor
- Maclaurin's
- expansions
- method
- model
- unit
- answer

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Taylor and Maclaurin's expansions without a clear example, method, or evidence.

Taylor and Maclaurin's expansions: Guided Activity

Guided activity for Taylor and Maclaurin's expansions:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Taylor and Maclaurin's expansions: Practice And Correction

Practice for Taylor and Maclaurin's expansions:

Main outcome: use Taylor and Maclaurin's expansion to solve problems about approximations, limits and integration. Extend the Maclaurin's expansion to Taylor series.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:

Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because $18 / 2 = 9$ km per hour.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to Taylor and Maclaurin's expansions and solve it.

Taylor and Maclaurin's expansions: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Taylor and Maclaurin's expansions.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Taylor and Maclaurin's expansions: Outcome Check

Outcome check for Taylor and Maclaurin's expansions: Use Taylor and Maclaurin's expansion to solve problems about approximations, limits and integration. Extend the Maclaurin's expansion to Taylor series.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Taylor and Maclaurin's expansions: Outcome Check

Outcome check for Taylor and Maclaurin's expansions: State the Taylor's formula for a function and the conditions for the expansion to be valid.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Integration: Lesson Opener

Learning focus: Integration

Country link: a Rwanda school, market, farming, building, transport, or data problem for Integration.

By the end of this topic, you should be able to:

- use integration as the inverse of differentiation and as the limit of a sum then apply it to find the area of plane surfaces, volumes of solids of revolution, lengths of curved lines
- define the differential of a function
- interpret geometrically the differential of a function
- list the differentiation formulas
- clarify the relationship between derivative and antiderivative of a function
- illustrate the use of basic integration formulas

Inquiry questions:

- How can integration help you solve a real problem in Core Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Integration: Learn

Integration teaches how two quantities change together at the same rate.

Core idea:

- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:

- Integration
- method
- model
- unit
- answer
- check

Integration: Worked Example

Worked example for Integration: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: $900 / 3 = 300$ francs.
2. Multiply by 5 books: $300 \times 5 = 1500$ francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Integration: Vocabulary And Study Skill

Use these words and study moves before you practise Integration. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Integration
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Integration without a clear example, method, or evidence.

Integration: Guided Activity

Guided activity for Integration:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Integration: Practice And Correction

Practice for Integration:

Main outcome: use integration as the inverse of differentiation and as the limit of a sum then apply it to find the area of plane surfaces, volumes of solids of revolution, lengths of curved lines.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:

Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because $18 / 2 = 9$ km per hour.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to integration and solve it.

Integration: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Integration.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Integration: Outcome Check

Outcome check for Integration: Use integration as the inverse of differentiation and as the limit of a sum then apply it to find the area of plane surfaces, volumes of solids of revolution, lengths of curved lines.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Integration: Outcome Check

Outcome check for Integration: Define the differential of a function.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Differential equations: Lesson Opener

Learning focus: Differential equations

Country link: a Rwanda school, market, farming, building, transport, or data problem for Differential equations.

By the end of this topic, you should be able to:

- use ordinary differential equations of first and second order to model and solve related problems in physics, economics, chemistry and biology
- extend the concepts of differentiation and integration to ordinary differential equations
- state the order and the degree of an ordinary differential equation
- express the auxiliary quadratic equation of a homogeneous linear differential equation of second order with constant coefficients
- predict the form of the particular solution of an
- determine whether an ordinary differential equation of first order is with separable variables, homogeneous or linear

Inquiry questions:

- How can differential equations help you solve a real problem in Core Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Differential equations: Learn

Differential equations teaches how two quantities change together at the same rate.

Core idea:

- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:

- Differential equations
- Differential
- equations
- method
- model
- unit
- answer
- check

Differential equations: Worked Example

Worked example for Differential equations: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: $900 / 3 = 300$ francs.
2. Multiply by 5 books: $300 \times 5 = 1500$ francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Differential equations: Vocabulary And Study Skill

Use these words and study moves before you practise Differential equations. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Differential equations
- Differential
- equations
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Differential equations without a clear example, method, or evidence.

Differential equations: Guided Activity

Guided activity for Differential equations:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Differential equations: Practice And Correction

Practice for Differential equations:

Main outcome: use ordinary differential equations of first and second order to model and solve related problems in physics, economics, chemistry and biology.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:

Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because $18 / 2 = 9$ km per hour.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to differential equations and solve it.

Differential equations: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Differential equations.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Differential equations: Outcome Check

Outcome check for Differential equations: Use ordinary differential equations of first and second order to model and solve related problems in physics, economics, chemistry and biology.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Differential equations: Outcome Check

Outcome check for Differential equations: Extend the concepts of differentiation and integration to ordinary differential equations.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Intersection and sum of subspaces: Lesson Opener

Learning focus: Intersection and sum of subspaces

Country link: a Rwanda school, market, farming, building, transport, or data problem for Intersection and sum of subspaces.

By the end of this topic, you should be able to:

- relate the sum and the intersection of subspaces of a vector space by the dimension formula
- define the intersection and the sum of subspaces of a vector space
- state the dimension formula
- list the conditions for a vector space to be qualified as direct sum of its subspaces
- perform the intersection and the addition of subspaces of a vector space
- determine a basis of the intersection and a basis of the sum of the subspaces of a vector space

Inquiry questions:

- How can intersection and sum of subspaces help you solve a real problem in Core Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Intersection and sum of subspaces: Learn

Intersection and sum of subspaces teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:

- Intersection and sum of subspaces
- Intersection
- subspaces
- method
- model
- unit
- answer
- check

Intersection and sum of subspaces: Worked Example

Worked example for Intersection and sum of subspaces: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:

1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Rwanda classroom, market, garden, transport, or home example.

Intersection and sum of subspaces: Vocabulary And Study Skill

Use these words and study moves before you practise Intersection and sum of subspaces. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Intersection and sum of subspaces
- Intersection
- subspaces
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Intersection and sum of subspaces without a clear example, method, or evidence.

Intersection and sum of subspaces: Guided Activity

Guided activity for Intersection and sum of subspaces:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Intersection and sum of subspaces: Practice And Correction

Practice for Intersection and sum of subspaces:

Main outcome: relate the sum and the intersection of subspaces of a vector space by the dimension formula.

1. Solve one intersection and sum of subspaces question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:

Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to intersection and sum of subspaces and solve it.

Intersection and sum of subspaces: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Intersection and sum of subspaces.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Intersection and sum of subspaces: Outcome Check

Outcome check for Intersection and sum of subspaces: Relate the sum and the intersection of subspaces of a vector space by the dimension formula.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Intersection and sum of subspaces: Outcome Check

Outcome check for Intersection and sum of subspaces: Define the intersection and the sum of subspaces of a vector space.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Transformation of matrices: Lesson Opener

Learning focus: Transformation of matrices

Country link: a Rwanda school, market, farming, building, transport, or data problem for Transformation of matrices.

By the end of this topic, you should be able to:

- transform matrices to an echelon form or to a diagonal matrix and use the results to solve simultaneous linear equations or to calculate the n th power of a matrix
- define the kernel, the image, the nullity and the rank of a linear transformation
- state the dimension formula for linear transformations
- carry out the elementary row operations on matrices
- define Eigen values and eigenvectors of a square matrix
- discuss the diagonalisation of a matrix of order 2

Inquiry questions:

- How can transformation of matrices help you solve a real problem in Core Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Transformation of matrices: Learn

Transformation of matrices teaches how two quantities change together at the same rate.

Core idea:

- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:

- Transformation of matrices
- Transformation
- matrices
- method
- model
- unit
- answer
- check

Transformation of matrices: Worked Example

Worked example for Transformation of matrices: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: $900 / 3 = 300$ francs.
2. Multiply by 5 books: $300 \times 5 = 1500$ francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Transformation of matrices: Vocabulary And Study Skill

Use these words and study moves before you practise Transformation of matrices. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Transformation of matrices
- Transformation
- matrices
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Transformation of matrices without a clear example, method, or evidence.

Transformation of matrices: Guided Activity

Guided activity for Transformation of matrices:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Transformation of matrices: Practice And Correction

Practice for Transformation of matrices:

Main outcome: transform matrices to an echelon form or to a diagonal matrix and use the results to solve simultaneous linear equations or to calculate the n th power of a matrix.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:

Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because $18 / 2 = 9$ km per hour.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to transformation of matrices and solve it.

Transformation of matrices: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Transformation of matrices.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Transformation of matrices: Outcome Check

Outcome check for Transformation of matrices: Transform matrices to an echelon form or to a diagonal matrix and use the results to solve simultaneous linear equations or to calculate the n th power of a matrix.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Transformation of matrices: Outcome Check

Outcome check for Transformation of matrices: Define the kernel, the image, the nullity and the rank of a linear transformation.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Conics: Lesson Opener

Learning focus: Conics

Country link: a Rwanda school, market, farming, building, transport, or data problem for Conics.

By the end of this topic, you should be able to:

- determine the characteristics and the graph of a conic given by its Cartesian, parametric or polar equation. Find the Cartesian, parametric and polar equations of a conic from its characteristics
- define geometrically a conic as the intersection of a plane and a cone and classify conics from the position of the intersecting plane
- express, in Cartesian form, the standard equation of a parabola, an ellipse and a hyperbola
- classify conics based on the value of the eccentricity
- find the characteristics of a conic given its equation (centre, axes, vertices, tangent, normal, shapes, asymptotes, foci, directrices, eccentricity, etc.)
- determine the Cartesian, parametric and polar equations of a conic from its characteristics

Inquiry questions:

- How can conics help you solve a real problem in Core Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Conics: Learn

Conics teaches properties of two-dimensional shapes.

Core idea:

- Shapes can be classified by sides, corners, lines of symmetry, and angles.
- A labelled sketch helps you compare shapes accurately.
- Use correct names such as triangle, rectangle, square, and circle.

Key words:

- Conics
- method
- model
- unit
- answer
- check

Conics: Worked Example

Worked example for Conics: Classify a shape with 4 equal sides and 4 right angles.

1. Four sides means it is a quadrilateral.
2. Equal sides and right angles match a square.
3. Check the name and properties.

Answer: The shape is a square.

Conics: Vocabulary And Study Skill

Use these words and study moves before you practise Conics. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Conics
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Conics without a clear example, method, or evidence.

Conics: Guided Activity

Guided activity for Conics:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Conics: Practice And Correction

Practice for Conics:

Main outcome: determine the characteristics and the graph of a conic given by its Cartesian, parametric or polar equation. Find the Cartesian, parametric and polar equations of a conic from its characteristics.

1. Draw a triangle, rectangle, and square.
2. Write two properties for each shape.
3. Circle the shape that has four equal sides and four right angles.

Answer check:

Answer: the square has four equal sides and four right angles.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to conics and solve it.

Conics: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Conics.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Conics: Outcome Check

Outcome check for Conics: Determine the characteristics and the graph of a conic given by its Cartesian, parametric or polar equation. Find the Cartesian, parametric and polar equations of a conic from its characteristics.

1. Draw a square and a rectangle.
2. Write two properties they share.
3. Write one property that is different.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Conics: Outcome Check

Outcome check for Conics: Define geometrically a conic as the intersection of a plane and a cone and classify conics from the position of the intersecting plane.

1. Draw a square and a rectangle.
2. Write two properties they share.
3. Write one property that is different.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Random variables: Lesson Opener

Learning focus: Random variables

Country link: a Rwanda school, market, farming, building, transport, or data problem for Random variables.

By the end of this topic, you should be able to:

- calculate and interpret the parameters of a random variable (discrete or continuous) including binomial and the Poisson distributions
- define a random variable
- identify whether a given random variable is discrete or continuous
- define the parameters of a discrete random variable
- learn in which situation the Binomial distribution applies and state its parameters
- use the concepts of statistics to compare frequency distribution to probability distribution

Inquiry questions:

- How can random variables help you solve a real problem in Core Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Random variables: Learn

Random variables teaches how to collect, organize, read, and explain information.

Core idea:

- Data must answer a clear question.
- Tables, tally charts, pictographs, and bar charts make data easier to read.
- A good conclusion says what the data shows.

Key words:

- Random variables
- Random
- variables
- method
- model
- unit
- answer

Random variables: Worked Example

Worked example for Random variables: A class records favourite fruits: mango 8, banana 12, pineapple 6. Which fruit is most popular?

1. Read the table values.
2. Compare 8, 12, and 6.
3. 12 is the greatest number.

Answer: Banana is most popular.

Random variables: Vocabulary And Study Skill

Use these words and study moves before you practise Random variables. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Random variables
- Random
- variables
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Random variables without a clear example, method, or evidence.

Random variables: Guided Activity

Guided activity for Random variables:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Random variables: Practice And Correction

Practice for Random variables:

Main outcome: calculate and interpret the parameters of a random variable (discrete or continuous) including binomial and the Poisson distributions.

1. Ask 10 learners a simple question and make a tally table.
2. Draw a bar chart from the tally table.
3. Write one conclusion from the data.

Answer check:

Answer check: totals in the tally table and bar chart should match.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to random variables and solve it.

Random variables: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Random variables.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Random variables: Outcome Check

Outcome check for Random variables: Calculate and interpret the parameters of a random variable (discrete or continuous) including binomial and the Poisson distributions.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Random variables: Outcome Check

Outcome check for Random variables: Define a random variable.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Complex numbers: Review Clinic 1

Use this review clinic to strengthen Core Mathematics without copying earlier answers.

1. Choose one idea from Complex numbers that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Logarithmic and exponential functions: Review Clinic 2

Use this review clinic to strengthen Core Mathematics without copying earlier answers.

1. Choose one idea from Logarithmic and exponential functions that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Taylor and Maclaurin's expansions: Review Clinic 3

Use this review clinic to strengthen Core Mathematics without copying earlier answers.

1. Choose one idea from Taylor and Maclaurin's expansions that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Integration: Review Clinic 4

Use this review clinic to strengthen Core Mathematics without copying earlier answers.

1. Choose one idea from Integration that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Glossary

Use this glossary to revise important words in Core Mathematics.

- Complex numbers: Explain this term using an example from the book, your school, or your community.
- Logarithmic and exponential functions: Explain this term using an example from the book, your school, or your community.
- Taylor and Maclaurin's expansions: Explain this term using an example from the book, your school, or your community.
- Integration: Explain this term using an example from the book, your school, or your community.
- Differential equations: Explain this term using an example from the book, your school, or your community.
- Intersection and sum of subspaces: Explain this term using an example from the book, your school, or your community.
- Transformation of matrices: Explain this term using an example from the book, your school, or your community.
- Conics: Explain this term using an example from the book, your school, or your community.
- Random variables: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Core Mathematics answer shows the model, method, units or labels, final answer, and a check. Use the worked examples and answer checks in each practice page to correct your work. When an answer is wrong, find the first step that changed the meaning and correct that step before trying a new question.

Final Project

Create a final project for Core Mathematics that shows what you can now do independently.

Your project must include:

1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.